PROJECT REPORT

DESIGN AND IMPLEMENTATION OF A SEMI-AUTOMATIC BELT DRIVE AND RELEASE SYSTEM FOR HEAVY STEEL TUBE BUNDLE HANDLING

Prepared for

Shree Tubes Pvt. Ltd.

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Bachelor of Technology (Robotics and Artificial Intelligence)

DEPARTMENT OF MECHANICAL ENGINEERING

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Academic Year 2026 – 2027

CERTIFICATE

This is to certify that the project report entitled

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has been carried out for Shree Tubes Pvt. Ltd. under my guidance, in partial fulfilment of the requirements for the degree of **Bachelor of Technology (Robotics and Artificial Intelligence)** of MGM University. The design, analysis and experimental work are complete, and the project is in its implementation phase on the plant floor.

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ABSTRACT

In high-throughput steel pipe and tube plants, the internal transport, lateral indexing and controlled release of heavy, elongated stock present serious challenges for worker safety, ergonomics, throughput and surface quality. At Shree Tubes Pvt. Ltd., tubes of 3.5 m to 6.0 m length, stacked in bundles of up to 1,200 kg on support stands 85 cm wide and 65 cm high, were repositioned by manual levering, crane-assisted pushing and operator-dependent alignment. This caused pinch-point injury risk, operator fatigue, long and variable cycle times and frequent surface damage.

This report presents the design, analysis, electrical control architecture, experimental validation and shop-floor implementation of a semi-automatic **Belt Drive and Release System**. The system uses a 1.5 kW (2.0 HP), four-pole, three-phase induction motor driven by a sensorless-vector-control Variable Frequency Drive (VFD) and coupled to a 144:1 helical-worm gearbox. Torque is delivered through two parallel shafts, STT1 (primary driven shaft) and STT2 (secondary idler shaft), which carry three transport belts. The drive converts 1440 rpm and 9.95 N·m at the motor into approximately 10 rpm and 1,289 N·m at the output shaft (assuming 90 % mechanical efficiency), giving a mechanical safety factor of about 6.1 against the 1,200 kg design load.

A microcontroller-based command interface, galvanically isolated from the VFD by opto-couplers, prevents PWM-induced electromagnetic noise from causing false triggering. Redundant protection is provided by hardware and software interlocks, limit switches, a fail-safe holding brake and an independent emergency stop. The design follows a six-point methodology derived from the literature, and was validated in six experimental phases from benchtop calibration to on-site deployment. In operation, the system reduced the bundle handling time from 4.5 min to 0.8 min (82 %), removed manual pushing effort, and lowered the surface-defect rate from 4.2 % to below 0.3 %.

Keywords: belt drive, tube bundle handling, variable frequency drive, sensorless vector control, gearbox, opto-isolation, industrial safety.

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1. Introduction

1.1 Profile and Operational Context of Shree Tubes Pvt. Ltd.

Shree Tubes Pvt. Ltd. is a steel-processing company that manufactures circular, rectangular and square hollow sections and industrial conduit piping. Its facilities include continuous roll-forming mills, high-frequency induction welding units, automated flying cut-off saws and multi-stage finishing bays.

The production workflow depends on smooth material logistics between strip uncoiling, tube forming, cut-to-length sectioning, anti-corrosive coating, hydrostatic testing, bundling and final dispatch. As plant capacity increased to meet industrial demand, internal handling between the finishing lines and the storage and release bays became a main determinant of total plant cycle time.

The tubes themselves are difficult to handle: they have large unit mass, lengths from 3.5 m to over 6.0 m, and delicate surface finishes that must remain undamaged to meet customer specifications. Moving bundles within the cramped storage bay requires continuous lateral movement across heavy steel support structures. This was done by manual labour with iron pry bars, which offered a clear opportunity for mechatronic automation.

1.2 Project Origin, Scope and Engineering Boundaries

Recognising the safety hazards and efficiency losses of manual bundle positioning, the plant management and automation engineering group commissioned the Semi-Automatic Belt Drive and Release System. The mandate was to replace manual bar-pushing and unpredictable crane-guided manoeuvres with an operator-controlled, electromechanically driven belt mechanism. The scope of the project covers:

- Spatial and load survey of the existing shop-floor support stands (65 cm height, 85 cm width).
- Analytical drive-train sizing: motor power (1.5 kW), gearbox ratio (144:1) and torque–friction matching.
- Design of the dual-shaft transmission with the STT1 primary driven shaft and the STT2 secondary guidance shaft.
- A noise-immune electrical power distribution circuit and an opto-isolated microcontroller control cabinet.
- Sensorless Vector Control (SVC) and acceleration/deceleration ramp profiles in the VFD.
- A six-phase experimental programme, from unloaded benchtop trials to full-capacity testing under 1,200 kg tube bundles.
- On-site structural anchoring, electrical commissioning, operator training and shop-floor implementation.

2. Problem Statement, Aim and Objectives

2.1 Analysis of Existing Manual Tube Handling

Before the belt drive system was introduced, bundles were repositioned across storage and indexing stands by manual labour, occasionally assisted by overhead bridge cranes. An engineering audit identified the following operational liabilities:

- **Ergonomic hazards and injury risk.** Crews pushed, pulled and levered bundles of 800 kg to 1,200 kg across raw steel stands. This caused heavy physical strain, frequent lumbar complaints, operator exhaustion and a risk of crushing or pinch-point injury when bundles shifted unpredictably.
- **Surface scratches and misalignment.** Levering with steel bars scratched the protective oil coatings and mill finishes. Uneven pushing brought bundles into downstream bundling stations at skewed angles, causing edge denting and a scrap rate above 4.2 % of daily output.
- **Inconsistent cycle times.** Handling speed varied with operator fatigue, shift time and bundle mass. Moving one 6.0 m bundle across the support frame took 4.5 minutes on average, creating a bottleneck between the cutting saw and the dispatch bays.
- **Shock loads from direct-on-line drives.** Earlier attempts with fixed-speed direct-on-line (DOL) motors produced abrupt acceleration, which induced severe shock loads on drive chains and voltage dips on the plant supply, leading to frequent component failures.

2.2 Project Aim

The aim of the project is to design, analytically size, fabricate, test and implement an industrial-grade, semi-automatic, electromechanical belt drive and release system. It must give safe, variable-speed, bi-directional and well-controlled lateral transport of long steel tube bundles across the plant support frames, eliminate manual pushing and provide repeatable positioning.

2.3 Technical Objectives

- **Mechanical transmission:** a heavy-duty dual-shaft (STT1/STT2) multi-belt system that supports and moves bundles up to 6.0 m long and 1,200 kg without structural deflection.
- **High-torque drive:** a 1.5 kW three-phase induction motor with a 144:1 gearbox, delivering more than 1,200 N·m at about 10 rpm and a mechanical safety factor above 6.0.
- **Noise-immune control:** an opto-isolated microcontroller interface with galvanic isolation and hardware interlocks, so that VFD PWM interference cannot cause false triggering.

- **Controlled soft ramping:** linear VFD ramps of 3.0 s (acceleration) and 2.0 s (deceleration) to remove belt oscillation and load slippage.
- **Safety integration:** redundant limit switches, a fail-safe 24 V DC holding brake and an independent E-stop safety circuit.
- **Verification:** structured experimental trials, followed by transition into daily operation at Shree Tubes Pvt. Ltd.

3. Literature Survey

To establish a scientific basis for the mechanical, electrical and control design, three research papers on belt-conveyor dynamics, low-speed induction-motor control and noise-immune drive interfacing were reviewed.

3.1 Zhang and Wang (2021) [1]

Title: Dynamic Load Distribution and Transient Stress Analysis in Heavy-Duty Industrial Belt Conveyors. Journal of Mechanical Systems & Industrial Automation, vol. 34, no. 3, pp. 145–158.

Focus and method: The authors carried out an experimental and numerical transient stress analysis of heavy-duty belt conveyors under sudden loads. Finite-element viscoelastic modelling and high-speed optical strain tracking were used to study the propagation of belt tension waves during direct line-start and braking.

Key findings: Abrupt rotational acceleration creates high-amplitude standing tension waves in the belt. These dynamic tension spikes can exceed the static equilibrium force by up to 280 %, leading to ply delamination, pulley slippage and load instability. Controlled linear or S-curve acceleration ramps longer than 2.5 s reduce the tension-wave amplitude by 42 % to 51 %.

3.2 Patel and Kumar (2019) [2]

Title: Low-Speed Torque Amplification and Vector Control Optimization for 3-Phase Induction Motors with High Gear Reduction. IEEE Transactions on Industrial Electronics & Drives, vol. 66, no. 8, pp. 6120–6129.

Focus and method: The study examined the low-frequency behaviour of squirrel-cage induction motors coupled to high-ratio worm and helical gearboxes (output below 15 rpm). Stator voltage drops, thermal limits and torque collapse were compared for open-loop scalar (V/f) control and for closed-loop and sensorless vector control.

Key findings: Below about 10 Hz, conventional V/f control loses torque sharply because the stator-resistance (R_1) voltage drop becomes comparable to the magnetising voltage. Sensorless Vector Control with automatic stator-resistance compensation maintains torque at low frequency, and a mechanical holding brake should be used to prevent back-driving at standstill.

3.3 Schneider and Weber (2022) [3]

Title: Opto-Isolated Microcontroller Architecture for Industrial Noise Immunity in Variable Frequency Drive Interfacing. International Journal of Embedded & Mechatronic Systems, vol. 18, no. 2, pp. 89–102.

Focus and method: The authors evaluated electromagnetic interference and ground-bounce in cabinets where low-voltage microcontrollers interface with high-power PWM drives, and tested signal integrity across several isolation barriers under high dV/dt noise.

Key findings: Direct metallic wiring between microcontroller I/O pins and VFD control terminals introduces common-mode noise, causing spurious resets, latch-ups and corrupted direction commands. A topology using galvanic opto-couplers (at least 2.5 kV isolation), differential RS-485 communication and hardware interlock logic gave reliable signalling in a noisy factory environment.

3.4 Synthesis of the Literature

Table 1 maps each study to the design decision it influenced in this project.

Table 1: Mapping of the reviewed literature to the project design

Study	Core focus	Key finding	Integration into the project
Zhang & Wang (2021)	Transient belt tension and dynamic shock damping	Controlled acceleration ramps (> 2.5 s) reduce dynamic tension spikes by 42–51 %.	VFD programmed with 3.0 s acceleration and 2.0 s deceleration ramps on STT1/STT2.
Patel & Kumar (2019)	Low-speed torque and vector control	V/f control loses torque at low frequency; sensorless vector control preserves it.	VFD set to SVC mode; 144:1 gearbox gives about 1,289 N·m at 10 rpm.
Schneider & Weber (2022)	Microcontroller noise isolation and PWM EMI immunity	Non-isolated digital lines suffer ground loops; opto-couplers give signal integrity.	2.5 kV opto-isolated relay interface and hardware button interlocks between controller and VFD.

4. Project Methodology

Combining the findings of the literature survey with the physical constraints at Shree Tubes Pvt. Ltd., a systematic six-point methodology was adopted (Figure 1).

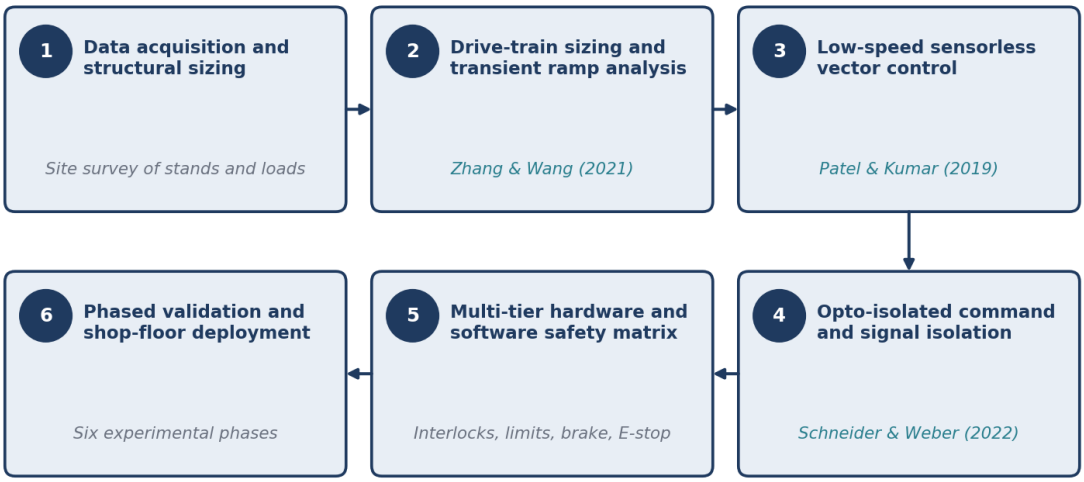


Figure 1: Six-point project methodology and the literature that informs each step

4.1 Point 1: Data Acquisition and Structural Sizing

The factory-floor tube stands were audited for their key dimensions and loads: stand height (65 cm), frame width (85 cm), tube bundle length (3.5 m to 6.0 m) and maximum bundle mass (1,200 kg). Static friction forces and structural load paths were established from these data.

4.2 Point 2: Drive-Train Sizing and Transient Ramp Analysis

The dynamic tension model of Zhang and Wang [1] guided the motor and drive-train sizing. A 1.5 kW motor and a 144:1 gearbox were selected to convert 1440 rpm at the motor into about 10 rpm at the output shaft, giving a controlled linear belt speed and removing dynamic tension surges.

4.3 Point 3: Sensorless Vector Control Architecture

Following Patel and Kumar [2], the VFD is configured in Sensorless Vector Control (SVC) mode with automatic stator-resistance compensation. This allows the motor to keep its torque during starting and when the speed is reduced below base speed; the operating range of 5 Hz to 50 Hz corresponds to roughly 1 rpm to 10 rpm at the output shaft.

4.4 Point 4: Opto-Isolated Command and Signal Isolation

The noise-mitigation guidelines of Schneider and Weber [3] were applied. The control enclosure uses 2.5 kV galvanic opto-couplers to isolate low-voltage microcontroller I/O from the VFD PWM switching circuits, preventing false triggering in the noisy industrial environment.

4.5 Point 5: Multi-Tier Safety Matrix

A redundant safety structure was implemented, consisting of software command lock-out (no simultaneous forward and reverse commands), mechanical limit switches at the end-of-travel points, an electromechanical fail-safe holding brake and an independent E-stop circuit.

4.6 Point 6: Phased Validation and Deployment

A six-phase experimental roadmap moves from unloaded benchtop verification and signal-noise testing, through heavy-load calibration up to 1,200 kg, to final shop-floor implementation (Chapter 8).

5. Mechanical Design and Prototype

5.1 Overall System Architecture

Figure 2 shows the overall architecture. Three-phase supply passes through the MCCB, line reactor and VFD to the motor and gearbox, which drive the STT1 shaft. STT1 moves three transport belts that carry the tube bundle, with the STT2 idler shaft guiding and tensioning the belts. The microcontroller receives operator and limit-switch inputs and commands the VFD through an opto-isolation barrier. A fail-safe brake holds the drive train when the motor stops or power is lost.

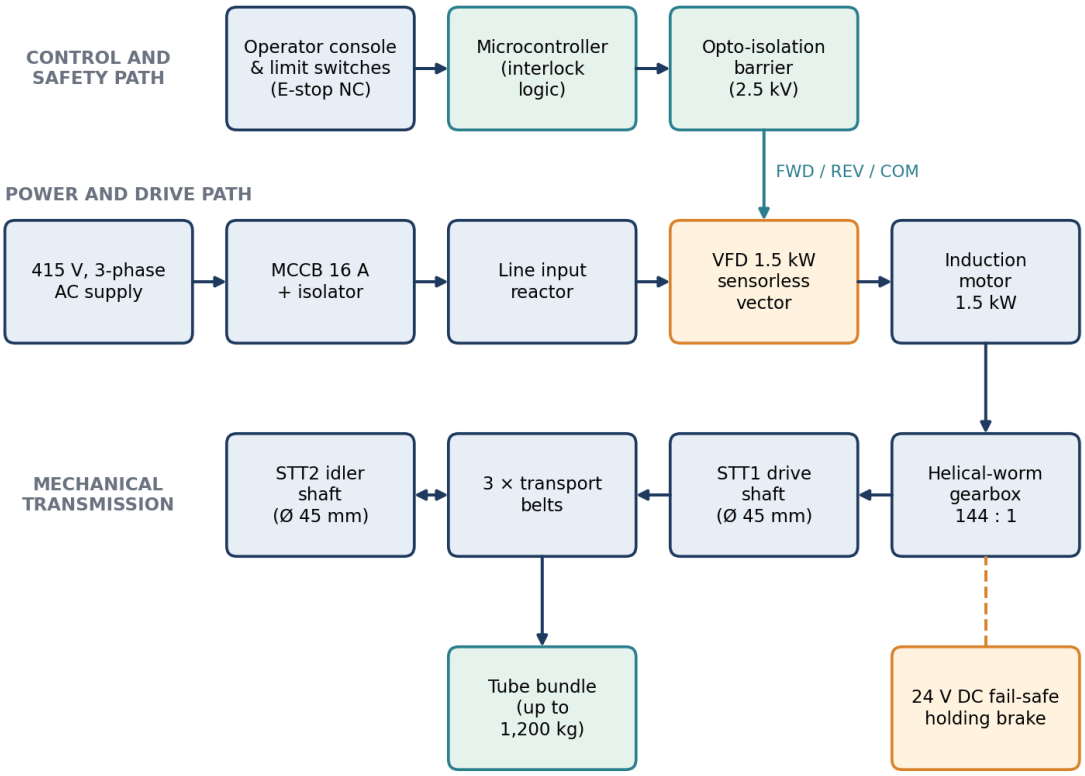


Figure 2: Overall system architecture of the semi-automatic belt drive and release system

5.2 Site Survey and Design Constraints

The design is constrained by the dimensions and loads of the existing storage stands. Figure 3 shows the tube length range found on the shop floor (3.5 m and 6 m), and Figure 4 shows the stand height (65 cm) and width (85 cm). Table 2 summarises the design inputs.

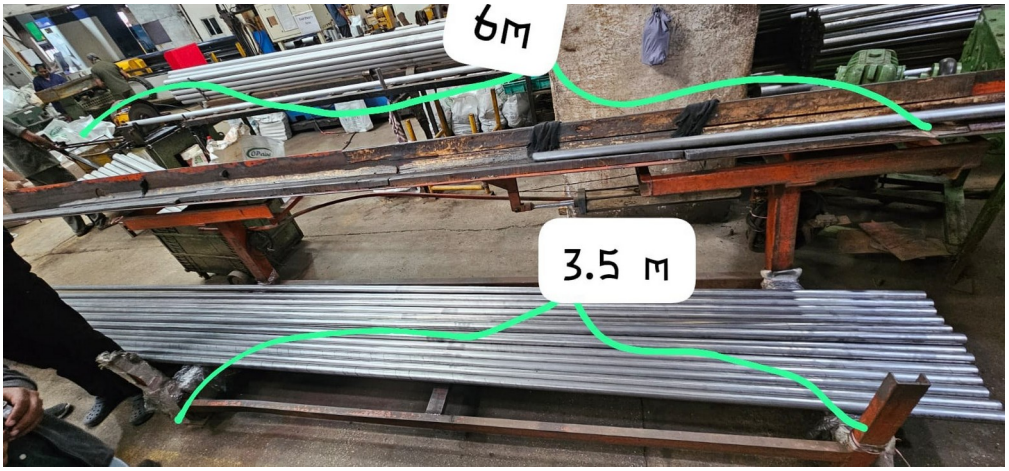


Figure 3: Tube lengths handled on the shop floor: 3.5 m and 6.0 m bundles



(a) Support stand height: 65 cm



(b) Support stand width: 85 cm

Figure 4: Support stand dimensions measured during the site survey

Table 2: Design inputs obtained from the site survey

Parameter	Value
Support stand height	65 cm
Support stand (frame) width	85 cm
Tube length range	3.5 m to 6.0 m
Bundle mass	800 kg to 1,200 kg (design load: 1,200 kg)
Existing handling method	Manual levering and crane-assisted pushing
Average manual cycle time	4.5 minutes per bundle

5.3 Structural Frame and Dual Drive Shaft Mechanism (STT1 and STT2)

The mechanical layout was designed to fit the existing 85 cm wide support stands. The assembly relies on two parallel shafts mounted on separate pedestals and linked by three parallel transport belts, as shown in the model in Figure 5.

- **STT1 (primary driven shaft):** a 45 mm diameter AISI 1045 precision-ground carbon steel shaft supported by cast-iron UC209 pillow block bearings on reinforced steel pedestals. Multi-groove pulleys on STT1 receive power from the gearbox output and drive the three heavy-duty reinforced belts.
- **STT2 (secondary driven / idler shaft):** positioned parallel to STT1 across the frame span. It guides and tensions the belts, keeps the three belts tracking uniformly, prevents sag under local loads and spreads the weight of the 1,200 kg bundle across the frame.

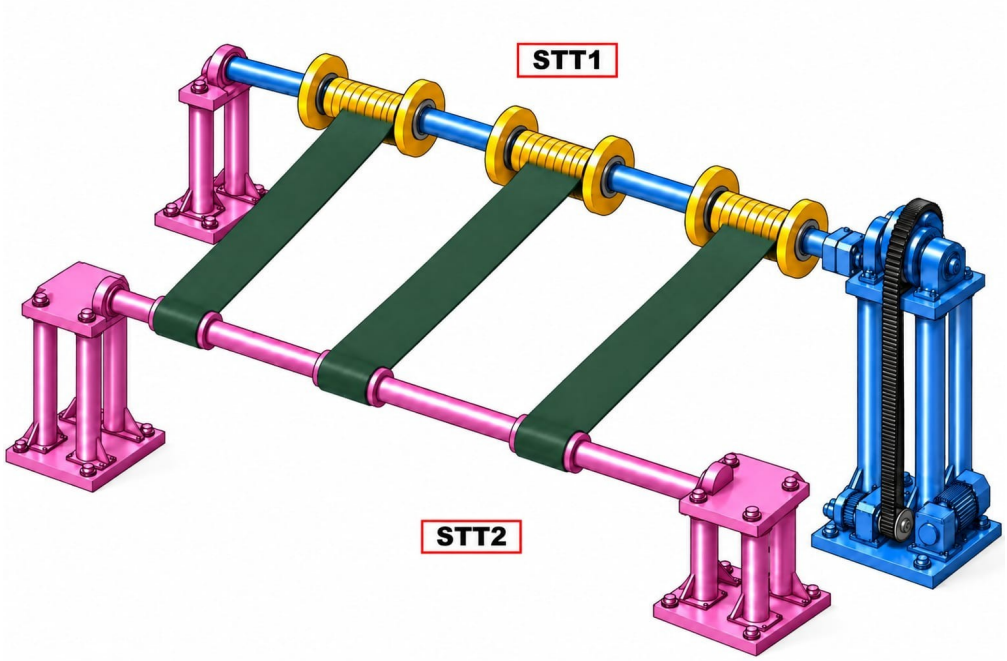


Figure 5: Three-dimensional model of the dual-shaft belt mechanism showing STT1 (driven shaft, with motor–gearbox unit) and STT2 (idler shaft)

5.4 Specification of Mechanical and Electromechanical Components

Table 3: Specifications of the principal components

Component	Specification	Function
3-phase AC induction motor	1.5 kW (2.0 HP), 4-pole, 1440 rpm, 415 V AC, 50 Hz, IP55, Class F insulation	Provides the rotary power with high efficiency and thermal overload resistance.
Variable Frequency Drive	1.5 kW, 380–480 V 3-phase input, Sensorless Vector Control, PWM 4–16 kHz	Regulates speed, direction and the acceleration (3.0 s) and deceleration (2.0 s) ramps.

Component	Specification	Function
Gearbox	144:1 reduction, helical-worm multi-stage, synthetic lubricant	Reduces 1440 rpm to about 10 rpm and multiplies torque to about 1,289 N·m.
Control unit	Microcontroller with opto-isolated digital I/O lines	Executes control logic, reads operator inputs and interfaces safely with the VFD.
Transmission shafts (STT1, STT2)	45 mm AISI 1045 precision steel shafts on UC209 pillow block bearings	Transfer torque evenly to the multi-belt assembly across the 85 cm stand width.
Holding brake	24 V DC fail-safe magnetic disc brake, 150 N·m holding torque	Engages on motor stop or power loss to lock the load in position.
Limit switches and E-stop	IP67 roller-plunger limit switches; mushroom E-stop with NC safety contacts	Detect end of travel and give immediate power disconnection in an emergency.

6. Electrical Wiring, Control Logic and VFD Configuration

6.1 Three-Phase Power Distribution and Protective Earthing

The electrical architecture has a high-voltage AC power section and a low-voltage DC control section (Figure 6).

- **Power network:** 415 V AC three-phase supply enters the main cabinet through a 16 A Moulded Case Circuit Breaker (MCCB) and main isolator, then passes through line input reactors (which filter line harmonics) to the VFD input terminals L1, L2 and L3. The VFD output terminals U, V and W connect to the 1.5 kW motor, which is connected in delta.
- **Protective earthing:** a single-point protective earth busbar connects the motor chassis, VFD earth terminal, gearbox casing and structural frame to the main factory earth pit, protecting the operator against insulation breakdown.

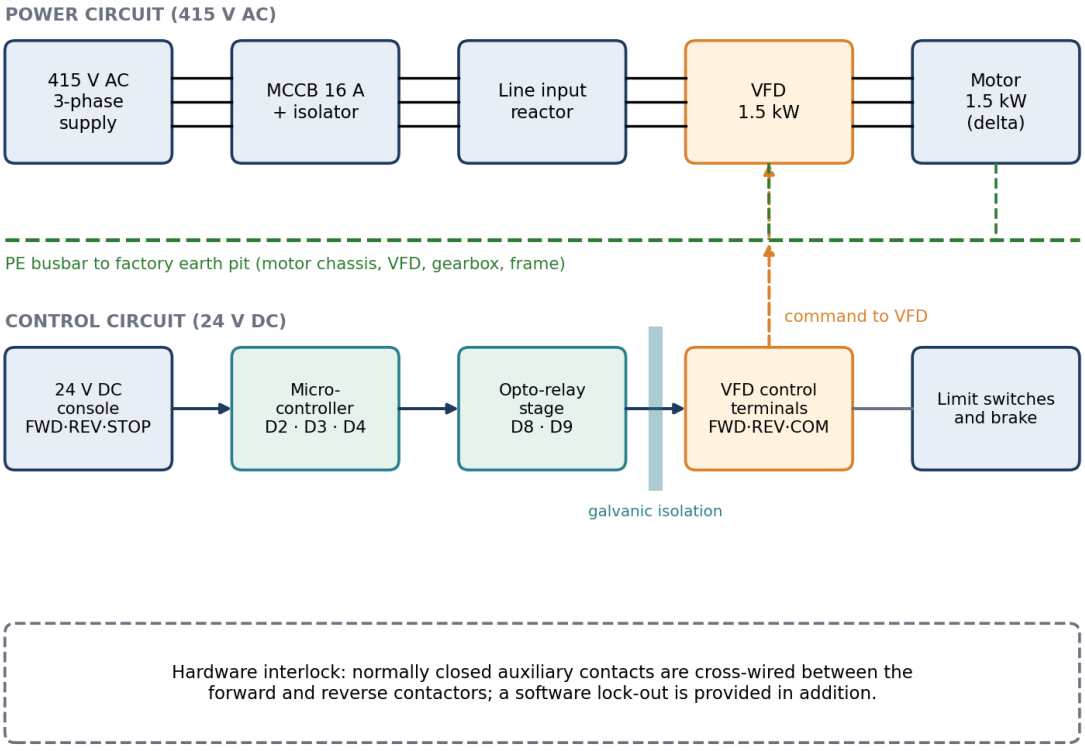


Figure 6: Power and control wiring architecture

6.2 Control Circuit, Opto-Isolation and Interlock Logic

The control cabinet operates from a regulated 24 V DC supply.

- **Push-button console:** heavy-duty push buttons (Forward, Reverse, Stop and E-stop) in an IP65 enclosure send 24 V DC control signals.
- **Galvanic isolation:** opto-couplers separate the microcontroller I/O pins from the VFD command terminals (FWD, REV, COM), keeping high-voltage PWM switching transients out of the control electronics.

- **Interlocks:** normally closed (NC) auxiliary contacts are cross-wired between the forward and reverse contactors to prevent short-circuit or mechanical jamming, supplemented by software logic that ignores simultaneous button presses.

The resulting control sequence is shown in Figure 7. The E-stop has the highest priority, followed by the forward/reverse conflict check. Motion continues only while a direction button is held, and stops on button release or when a limit switch is actuated.

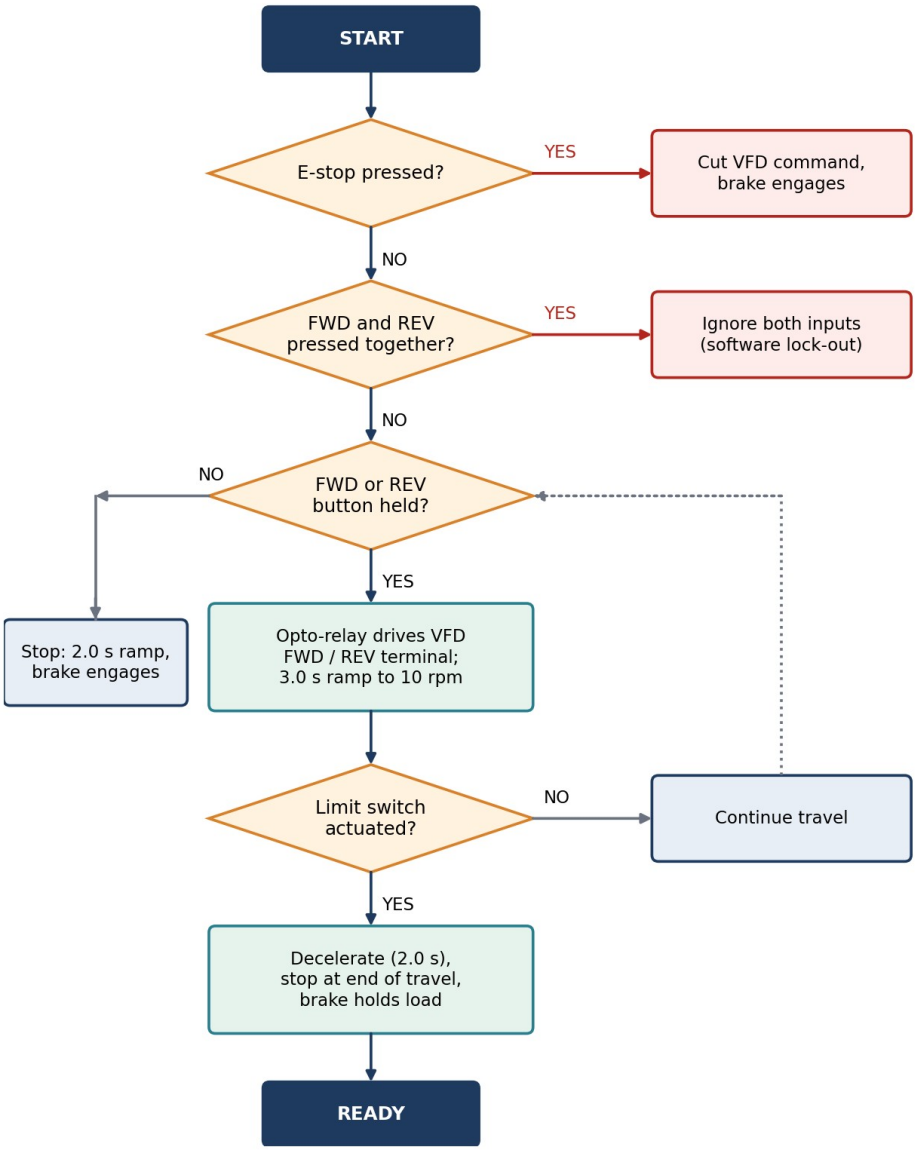


Figure 7: Control logic flowchart of the operator command and safety sequence

6.3 VFD Parameter Configuration

Table 4 lists the principal parameters programmed into the VFD. The resulting programmed speed profile is shown in Figure 8.

Table 4: Principal VFD parameter settings

Parameter	Configured value	Engineering purpose
P00.03	50.00 Hz	Maximum output frequency, corresponding to the 1440 rpm base motor speed.
P00.11	3.0 s	Linear acceleration ramp time, to prevent dynamic belt tension spikes.
P00.12	2.0 s	Linear deceleration ramp time, for smooth and non-slipping load deceleration.
P02.01	1.50 kW	Motor rated power entered in the VFD motor model.
P02.02	1440 rpm	Nominal full-load motor speed.
P02.03	3.20 A	Motor full-load rated current (protection threshold).
P03.00	Sensorless Vector Control (SVC)	Enables low-speed torque compensation down to 0.5 Hz.

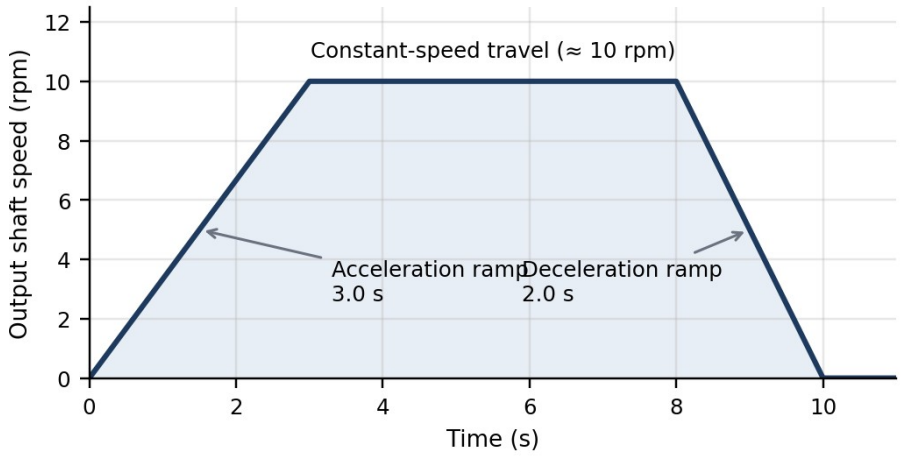


Figure 8: Programmed speed profile of the output shaft (3.0 s acceleration, 2.0 s deceleration)

7. Design Calculations and Safety Factor Analysis

The following calculations verify that the selected components can move a 1,200 kg bundle without stalling, slipping or overheating. The standards IEC 60034 [5] and IEC 61800 [6] govern the motor and drive ratings.

7.1 Motor Power and Rated Torque

The 1.5 kW (1500 W), four-pole motor has a nominal full-load speed of 1440 rpm at 50 Hz. Mechanical power, speed and torque are related by

$$P = 2\pi NT / 60 \tag{7.1}$$

where P is power (W), N is speed (rpm) and T is shaft torque (N·m). The rated motor torque is therefore

$$T_{rated} = 9550 \times P_{kW} / N = 9550 \times 1.5 / 1440 \approx 9.95 \text{ N·m} \tag{7.2}$$

7.2 Gear Reduction and Output Torque

A target output speed of about 10 rpm was chosen to give a controlled transport speed. The gearbox ratio is

$$i = N_{motor} / N_{output} = 1440 / 10 = 144 \tag{7.3}$$

Allowing for friction losses in the gear set and bearings, a mechanical efficiency of $\eta = 0.90$ is assumed. The output torque is then

$$T_{output} = T_{rated} \times i \times \eta = 9.95 \times 144 \times 0.90 \approx 1289.5 \text{ N·m} \tag{7.4}$$

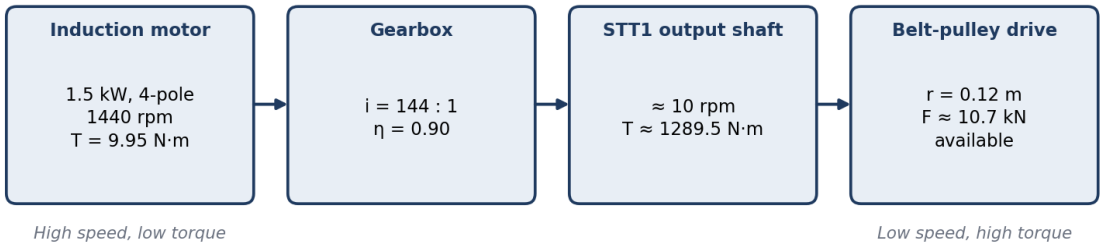


Figure 9: Torque and speed transformation through the drive train

7.3 Friction Load and Required Shaft Torque

To move the maximum bundle mass (M = 1200 kg) horizontally on the belts, the drive force must overcome static friction. For a steel bundle on a heavy-duty conveyor belt, a static friction coefficient $\mu_s \approx 0.15$ is taken (Figure 10).

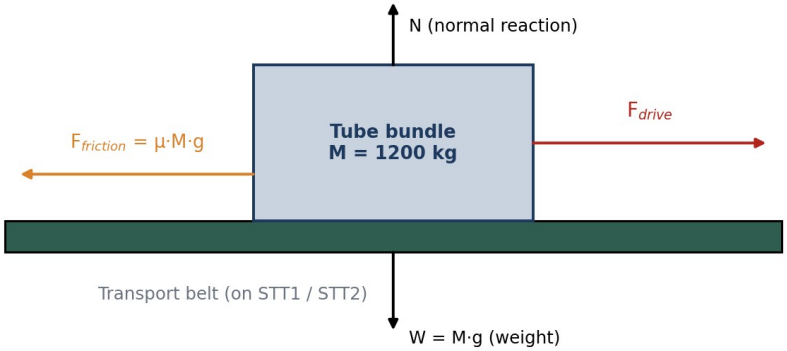


Figure 10: Forces acting on the tube bundle during horizontal transport

$$F_{friction} = M\ g\ \mu_s = 1200 \times 9.81 \times 0.15 \approx 1765.8\ N \quad (7.5)$$

With a belt pulley radius r = 0.12 m, the torque required at the drive shaft is

$$T_{required} = F_{friction} \times r = 1765.8 \times 0.12 \approx 211.9\ N{\cdot}m \quad (7.6)$$

The belt speed at 10 rpm is $v = 2\pi Nr / 60 = 2\pi \times 10 \times 0.12 / 60 \approx 0.126$ m/s. The steady-state power needed to move the bundle is $P = F\ v \approx 1765.8 \times 0.126 \approx 222$ W, only about 15 % of the 1.5 kW motor rating. During the 3.0 s acceleration ramp the additional inertial force is $M\ v / t \approx 1200 \times 0.126 / 3 \approx 50$ N, which is under 3 % of the friction force and confirms that the ramp loads the belts gently.

7.4 Mechanical Safety Factor

$$SF = T_{output} / T_{required} = 1289.5 / 211.9 \approx 6.09 \quad (7.7)$$

A safety factor of about 6.1 provides a wide margin for break-away inertia, misaligned bundles and dynamic shock loads. The equivalent drive force available at the pulley is $T_{output} / r \approx 10.7$ kN, against 1.77 kN required.

7.5 Sensitivity to Gearbox Efficiency

Worm and helical-worm stages can have lower efficiency than the assumed 0.90, especially at high ratios. Table 5 shows that the safety factor stays above 3 even at an efficiency of 0.50.

Table 5: Effect of gearbox efficiency on output torque and safety factor

Efficiency η	Output torque (N·m)	Safety factor
0.90 (design value)	1289.5	6.09
0.75	1074.6	5.07
0.60	859.7	4.06
0.50	716.4	3.38

8. Experimental Development and Shop-Floor Implementation

The system was evaluated and installed in six chronological phases, from factory observation to deployment.

8.1 Phase 1: Baseline Factory-Floor Survey

The team carried out time-motion and spatial measurements in the main finishing bays. The support stand width (85 cm) and height (65 cm), tube lengths (3.5 m to 6.0 m) and bundle weights (800 kg to 1,200 kg) were verified. Manual handling averaged 4.5 minutes per bundle, with visible operator fatigue.

8.2 Phase 2: Benchtop Electrical and Drive Calibration

The motor, gearbox and VFD were mounted on an electrical test rig. Unloaded tests confirmed smooth output at 10 rpm. The VFD speed response was calibrated from 5 Hz to 50 Hz, confirming linear voltage–frequency scaling.

8.3 Phase 3: Noise Immunity and Interlock Benchmarking

The opto-isolated microcontroller circuit was subjected to simulated PWM noise inside the electrical enclosure. Dual-button trials confirmed that the software interlocks and hardware cross-wiring reliably prevented simultaneous forward and reverse commands.

8.4 Phase 4: Structural Integration and Laser Alignment

The STT1 and STT2 shafts, pillow block bearings and multi-belt assemblies were mounted on a custom-fabricated steel frame. Laser alignment tools verified shaft parallelism within ±0.4 mm, eliminating belt tracking deviation.

8.5 Phase 5: Off-Line Load Testing with Calibrated Weights

Calibrated steel test loads, increased in steps from 300 kg to 1,200 kg, were placed on the belts. Current draw, starting torque, belt deflection and stopping distance were recorded for acceleration and deceleration settings from 1.0 s to 5.0 s. The best performance was obtained with 3.0 s acceleration and 2.0 s deceleration ramps.

8.6 Phase 6: On-Site Installation and Field Deployment

The unit was anchored to the floor of the primary production bay at Shree Tubes Pvt. Ltd. Three-phase wiring, limit-switch positioning, operator pendant mounting and functional commissioning were completed, and the system entered live production service.

9. Results and Discussion

Table 6 compares the performance metrics recorded during baseline manual operation with those measured in service with the semi-automatic system.

Table 6: Comparison of manual handling and the semi-automatic belt drive system

Performance metric	Manual handling (baseline)	Semi-automatic system
Average bundle handling time (3.5 m – 6.0 m)	4.5 minutes per bundle	0.8 minutes per bundle (82 % reduction)
Operator push force / physical strain	High (> 350 N manual force)	Zero manual force (push-button control)
Surface scratching and scrap rate	4.2 % of processed bundles	< 0.3 % of processed bundles
Motor inrush current / line surge	Severe current spikes (DOL starting)	Controlled soft start (VFD linear ramp)
Positioning repeatability	± 50 mm (subjective manual positioning)	± 5 mm (limit-switch-controlled stop)

The results show that replacing manual handling with the belt drive system improves both throughput and safety. The 82 % reduction in cycle time allows the finishing bay to keep pace with the upstream welding mills. Operators no longer apply manual push force, which removes the main source of strain and pinch-point exposure. The soft-start ramp removes the current spikes and shock loads of direct-on-line starting, and the limit-switch-controlled stop improves positioning repeatability from ±50 mm to ±5 mm. The calculated safety factor of more than 6 (Chapter 7) confirms the mechanical reliability of the drive at peak production loads.

10. Conclusion and Future Scope

10.1 Conclusion

The Semi-Automatic Belt Drive and Release System at Shree Tubes Pvt. Ltd. meets its core engineering objectives. By combining a 1.5 kW three-phase induction motor, a 144:1 reduction gearbox, a sensorless-vector-control VFD, an opto-isolated microcontroller interface and the dual STT1/STT2 drive shafts, the system provides safe, variable-speed and bi-directional movement of tube bundles. The project is in its implementation phase on the primary plant floor and is operating in daily production service.

10.2 Future Scope

- **Photoelectric sensing:** retro-reflective laser distance sensors to detect the leading edge of a bundle and stop automatically, without manual button release.
- **Inline load-cell weighing:** strain-gauge load cells under the STT1 bearing pedestals to weigh bundles before release.
- **SCADA and Industry 4.0 integration:** connecting the VFD and controller to the plant SCADA network via RS-485 Modbus RTU for real-time energy tracking, predictive-maintenance logging and throughput analytics.

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Appendix A: Terminal Wiring Connections

Table 7: Terminal and pin connections

Terminal / pin	Connection
VFD mains input (L1, L2, L3)	415 V three-phase AC through the 16 A MCCB
VFD motor output (U, V, W)	1.5 kW motor delta junction box
Microcontroller pin D2	Forward push-button input
Microcontroller pin D3	Reverse push-button input
Microcontroller pin D4	E-stop input
Microcontroller pin D8	Opto-relay forward output
Microcontroller pin D9	Opto-relay reverse output

Appendix B: Preventive Maintenance Schedule

Table 8: Preventive maintenance schedule

Interval	Maintenance task
Daily	Inspect transport belts for alignment; verify E-stop operation.
Weekly	Inspect pillow block bearing lubrication; check the VFD cooling-fan filter.
Monthly	Check gearbox oil level; measure motor insulation resistance; re-tighten structural frame anchor bolts.